

DEPLOYMENT PLAN

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1. Introduction

1.1 Purpose

This deployment plan outlines the strategy, infrastructure, and operational procedures required for deploying MediSync+. The system is an AI-powered post-discharge risk detection platform designed to bridge the gap between hospital discharge and follow-up care.

1.2 Objectives

The objectives of this deployment plan are to:

- Ensure reliable and secure deployment
- Minimize downtime using Blue-Green deployment
- Maintain compliance with PDPA
- Enable scalability across healthcare providers
- Establish monitoring and rollback procedures

1.3 Scope

This deployment plan covers:

- Android application deployment
 - Backend API deployment
 - AI service deployment
 - Database configuration
 - Security and compliance
 - Monitoring and recovery
-

2. System Overview

MediSync+ is an AI-assisted healthcare platform that detects patient deterioration after discharge.

Core Features

- Medication adherence tracking
- Symptom monitoring
- AI-powered risk detection

- Chat-based virtual nurse

Components

- Android mobile app (offline-first)
 - Backend services (Node.js + PostgreSQL)
 - Multi-agent AI system
 - Monitoring infrastructure
-

3. Deployment Architecture

3.1 Environment Configuration

3.1.1 Development Environment

The development environment is used for:

- Feature development
- Internal testing
- AI prompt experimentation
- Integration validation

Characteristics:

- Isolated local and cloud-based setup
- Synthetic and anonymized datasets
- Non-production credentials

3.1.2 Staging Environment

The staging environment replicates the production environment and is used for:

- User Acceptance Testing (UAT)
- Clinical workflow validation
- Security testing
- Performance benchmarking

Characteristics:

- Production-like infrastructure
- De-identified testing data

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- Controlled user access

3.1.3 Production Environment

The production environment hosts the live MediSync+ platform.

Characteristics:

- High availability configuration
- Architecture is designed to support future regional redundancy.
- Secure cloud infrastructure
- Planned monitoring and automated backup mechanisms.

3.2 Core Infrastructure Components

Frontend Application

The frontend application provides:

- Organized Recovery Roadmaps
- Medication reminders
- Symptom logging interfaces
- Follow-up scheduling

Supported platforms:

- Mobile devices

Backend Orchestrator

The backend orchestrator is responsible for:

- API management
- Workflow coordination
- Authentication and authorization
- Notification management
- Integration with AI services

AI Service Layer

The AI layer supports:

- Clinical instruction simplification
- Medical terminology translation
- Conversational assistance
- Recovery risk monitoring

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The AI system functions as a clinical support tool and does not independently diagnose medical conditions.

Database Infrastructure

Database systems include:

- PostgreSQL for structured patient workflow data
- Optional NoSQL databases for conversational logs and analytics

Features:

- Encrypted storage
- Role-based access control
- Audit logging
- Backup and recovery support

Queue Management System

Redis or RabbitMQ will be used for asynchronous task processing, including:

- Notification delivery
 - Scheduled reminders
 - Recovery trend generation
 - Background AI processing
-

4. Deployment Strategy

4.1 Containerization

All services will be containerized using Docker to ensure:

- Environment consistency
- Scalability
- Simplified deployment
- Service isolation

4.2 Continuous Integration and Continuous Deployment (CI/CD)

A CI/CD pipeline will be implemented using GitHub Actions or GitLab CI/CD.

Pipeline stages include:

1. Source code validation
2. Automated testing

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3. Security scanning
4. Integration testing
5. Deployment to staging
6. User Acceptance Testing (UAT)
7. Controlled production deployment

4.3 Deployment Methodology

A Blue-Green deployment strategy will be used to reduce deployment downtime and risk.

Deployment process:

1. Deploy updated version to parallel environment
 2. Conduct validation and monitoring
 3. Gradually switch production traffic
 4. Maintain rollback capability if failures occur
-

5. Data Management Plan

5.1 Database Migration

Database schema updates will be managed using version-controlled migration tools such as:

- Flyway
- Liquibase
- Alembic

5.2 Data Access Control

Patient Access

Patients may:

- View approved recovery instructions
- Submit symptom logs
- Record medication adherence
- Provide recovery feedback

Patients are not permitted to modify clinical instructions.

Staff Access

Healthcare staff may:

- Monitor patient recovery status

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- Review escalation alerts
- Access analytics dashboards
- Manage follow-up workflows

Access privileges are granted according to the principle of least privilege.

5.3 Backup and Recovery

The backup strategy includes:

- Daily encrypted backups
- Point-in-Time Recovery (PITR)
- Disaster recovery procedures
- Periodic restoration testing

5.4 Compliance

Data handling practices are designed to align with:

- Malaysian Personal Data Protection Act (PDPA)
 - Hospital cybersecurity requirements
 - Healthcare data governance standards
-

6. Security Plan

6.1 Authentication

Authentication mechanisms include:

- JWT or OAuth2 authentication
- Multi-factor authentication (MFA) for healthcare staff
- Session timeout controls

6.2 Authorization

Role-Based Access Control (RBAC) policies ensure:

- Restricted patient access
- Controlled staff permissions
- Administrative privilege management

6.3 Encryption

Security measures include:

- TLS encryption for data in transit

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- AES-256 encryption for stored data
- Secure API communication channels

6.4 Audit Logging

System activities recorded include:

- Authentication events
 - Patient data access
 - Administrative actions
 - Escalation reviews
-

7. AI Deployment Considerations

7.1 Instruction Organizer

The Instruction Organizer converts complex discharge instructions into structured and patient-friendly recovery plans.

Functions include:

- Clinical language simplification
- Chronological task organization
- Medication scheduling support
- Recovery adherence assistance

7.2 Risk Monitoring Engine

The Risk Monitoring Engine tracks patient recovery trends using:

- Pain level monitoring
- Mobility tracking
- Medication adherence
- Daily symptom logs

Potentially concerning recovery patterns trigger alerts for healthcare staff review.

The system supports clinical decision-making and does not replace medical professionals.

8. Monitoring and Observability

8.1 System Monitoring

Monitored metrics include:

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- API response latency
- Error rates
- Queue processing performance
- AI service response times

8.2 Operational Monitoring

Operational metrics include:

- Recovery roadmap completion rates
- Symptom logging consistency
- Risk alert accuracy
- Escalation response times

8.3 Observability Tools

Monitoring infrastructure includes:

- ELK Stack or Grafana Loki for centralized logging
 - Prometheus and Grafana for metrics visualization
 - Real-time alerting systems
-

9. Testing Plan

9.1 Technical Testing

Testing activities include:

- Unit testing
- Integration testing
- API testing
- Security testing
- Load testing

9.2 Clinical Validation

Healthcare professionals will review AI-generated outputs to:

- Validate instruction clarity
- Verify workflow consistency
- Minimize clinically significant misunderstandings

9.3 Performance Testing

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Peak discharge simulations will be conducted to verify system scalability and stability.

10. Rollout Plan

Phase 1 – Internal Pilot

- Staff-only deployment
- Workflow validation
- Internal feedback collection

Phase 2 – Limited Patient Deployment

- Controlled rollout to selected patients
- Monitoring of recovery workflows
- Feedback-driven improvements

Phase 3 – Expanded Deployment

- Deployment to additional healthcare providers
 - Activation of broader monitoring capabilities
 - Continuous operational optimization
-

11. Rollback and Recovery Plan

11.1 Version Rollback

The CI/CD pipeline supports rapid rollback to previously stable application versions.

11.2 Feature Flags

Feature flags allow selective disabling of:

- AI services
- Monitoring modules
- Experimental features

without affecting core platform functionality.

11.3 Database Recovery

Recovery mechanisms include:

- Database snapshots
- Backup restoration procedures
- Point-in-Time Recovery (PITR)

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12. Post-Deployment Maintenance

12.1 Prompt Optimization

AI prompts and workflows will be continuously refined using:

- Patient feedback
- Clinical feedback
- System analytics

12.2 Security Maintenance

Maintenance activities include:

- Vulnerability patching
- Dependency updates
- Security audits
- Penetration testing

12.3 Model Governance

Periodic evaluations will monitor:

- Hallucination rates
- Unsafe outputs
- Bias risks
- Model performance drift

12.4 Continuous Improvement

Future improvements will focus on:

- Deploy self-hosted AI infrastructure
- Integrate wearable devices
- Develop a healthcare provider dashboard
- Create a MediSync+ Server

13. Conclusion

This deployment plan establishes the technical, operational, and security framework required for the successful deployment of MediSync+. Through structured deployment procedures, AI governance, monitoring infrastructure, and healthcare-focused security controls, the platform

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aims to provide reliable post-discharge risk detection while maintaining scalability, compliance, and patient safety.